MECHANISM 05 OF 26 · DATA CENTERS AND HARDWARE

Physical chip inventory inspection

What the mechanism checks, where it appears in proposals and literature, and the tradeoff between evidence and invasive access.

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What this mechanism does

Physical inspection places a trusted observer at a facility to compare the real hardware with declared inventory. Inspectors may count accelerator modules, read or challenge unique identifiers, examine seals, and sample racks against purchase and registry records. The central claim is that certain chips were present, absent, or altered at a particular time. This is stronger than an unverified owner statement, but it is still a snapshot. It does not by itself say which workloads ran before or after the visit.

Sources, proposals, and literature

Pacing proposals set out interventions; verification frameworks develop oversight methods; literature includes safety and auditing studies. A citation does not imply a commitment to deploy this exact method.

Mechanism at a glance

Field	Summary
What it does	Inspect declared sites and sample chip identities and counts against a registry.
Checks and why	Installed accelerators and identifiers; to detect missing, substituted or hidden chips.
Pacing policy families	Compute limits
Direct proposals	Scher & Thiergart, Mechanisms to Verify International Agreements About AI Development
Broader verification frameworks	CNAS, Secure, Governable Chips
Literature	Sastry et al., Computing Power and the Governance of Artificial Intelligence
Technical feasibility	**High** — Counting and sampling are available now; scale and anti-tamper work raise cost.
Policy and strategy feasibility	**Low** — Intrusive access to private or sovereign facilities is difficult to negotiate.

How it works as verification

Inspection design depends on the threat model. At a cooperative cloud facility, auditors may need only to verify that a sampled set of serials is present and that records match rack positions. In a low-trust international agreement, inspectors may need more independent control over sampling, access to suspect areas, and checks for disguised hardware or tampered monitors. The MIRI verification survey presents data-center inspections as a near-term approach to locating compute, paired with a chip registry and, where needed, continuous monitoring.

Several practical details matter. The inspector must know what “covered chip” means, whether it can inspect packaged spares, and whether there are areas excluded for security or customer confidentiality. A facility may use mixed AI and non-AI equipment in the same hall. Opening racks can disrupt service or expose private architecture. Inspections can therefore combine record review, external visual checks, samples of powered and unpowered devices, and a secure method to attest device identity without reading customer data. The sampling plan should be agreed in advance but the actual sample should remain unpredictable.

The CSET report on tracking AI chips compares physical inspection with other location-verification methods. It finds inspections useful for investigating leads and complementary to automated location checks, while recognizing their cost and limited frequency. That framing is important: inspection is most valuable as an independent anchor for records and sensors, not as a continuous account of every chip.

An inspection begins before the site visit. The verifier reconciles reported purchases, transfers, repairs, and retirements to establish an expected population. It selects a random sample and some risk-based targets. At the site, it checks facility boundaries, inventories, sealed storage, and selected racks. It compares observed identifiers with registry entries and records any inaccessible area or device. Afterward, the verifier resolves discrepancies with independent documents, not just an amended inventory supplied by the operator.

For pacing, the inspection's role changes with the policy. A rule limiting total advanced compute needs counts and evidence that chips have not been shifted to an unregistered site. A rule forbidding large training may use the inventory only to identify facilities capable of violating the rule, then apply workload monitoring there. A rule restricting exports may care more about location than activity. The inspection report should therefore name the claim it tests. A “pass” on chip count must never be presented as proof that no prohibited run occurred.

Frequency can be risk based. Sites holding a small number of older chips may require only occasional review. A frontier-scale cluster near a policy threshold may warrant surprise visits or continuous portal monitoring. The protocol should also cover what happens when a facility declines access: does the refusal count as a breach, trigger a narrower license, or lead to a negotiated follow-up? Without a predefined response, inspection rights can be ineffective in the very cases where suspicion is highest.

Inspection can protect confidentiality through managed access, trained personnel, separated routes, and reporting that records only necessary facts. However, such limits cannot be allowed to make the check purely ceremonial. The verifier needs enough independent choice of samples and enough visibility to assess undeclared hardware. The tension between intrusive access and credible assurance should be described openly rather than hidden behind “third-party audit.”

A trial inspection should include a declared site with a mix of installed accelerators, spares, and equipment under repair. Inspectors choose serials after arrival, compare them with manufacturer and registry records, and record the time and access needed. A second, unannounced visit tests whether the facility can sustain the process without advance preparation. The report should distinguish inaccessible areas from areas examined and explain whether the sample was large enough for the policy claim.

Verification versus invasiveness

What the evidence can establish

CHECKABLE CLAIM
Whether a facility physically holds the number and type of covered accelerator chips it declares.

BOUNDARY / FALSE ASSURANCE
A truthful count at inspection time does not establish continuous location or workload history. A facility can present its declared racks while borrowing remote chips for a covered training run.

Access and affected parties

VERIFIER TOUCHES
Inspectors need site entry, rack access, serial or package inspection, procurement records and potentially maintenance windows to confirm installed hardware.

WHO BEARS THE BURDEN
Data centers and labs: facility security, uptime and staff time. Other tenants: co-location confidentiality. Inspectors: physical safety and chain of custody.

Vulnerabilities and evasion

ESPIONAGE, SABOTAGE AND PRIVACY EXPOSURE
Site access exposes layouts, security procedures, tenants and hardware placement. A malicious or compromised inspector could plant devices, damage equipment or collect intelligence.

HOW TO EVADE THIS CHECK
Move hardware before scheduled inspections, conceal it in unvisited rooms, use remote capacity, substitute lookalike devices or falsify labels. Sampling can miss small hidden clusters.

Incentives and a narrower design

INCENTIVE CREATED BY VERIFICATION
Frequent inspections increase downtime and physical security costs, and may push capacity into smaller or foreign sites that are harder to visit.

PROPORTIONATE IMPLEMENTATION
Use risk-based surprise visits, read-only scans and chain-of-custody sampling; reconcile to purchases, network topology and remote attestation. Scope inspectors away from unrelated tenants.

Scope: The analysis concerns the verification method. A pacing policy can fail for additional reasons even when its measurement is accurate. These are analytical failure scenarios, not claims that a named organization has done them.

Further limitations in the source brief

Snapshot problem. Chips can be moved after an inspection. If visits are announced far in advance, a violator can temporarily relocate equipment or alter displayed records. Random timing, seals, cameras, portal logs, and remote location checks reduce the opportunity but create new access and trust requirements. Inspection frequency must be related to how quickly a prohibited activity could occur.

Circumvention. An operator can hide a rack, present substitute devices, copy visible identifiers, keep spare chips off-site, or deny access to a sensitive room. A more capable adversary might fabricate matching paperwork and logs. Inspectors need anti-tamper procedures, the right to sample unexpected areas within a defined scope, and a way to compare serials to manufacturer records. Even a successful inspection of declared facilities cannot detect an entirely undeclared site without outside intelligence or supply-chain evidence.

Adoption resistance. Data centers contain customer secrets, security controls, and valuable equipment. Operators may fear disruption and exposure of network layouts. Governments may resist foreign inspectors at strategic sites. Inspectors themselves need specialized training and secure handling of what they see. The cost grows with the number and geographic spread of sites. CSET's location-verification assessment argues that automated ping-based checks can scale more cheaply, with physical inspections used to investigate alerts.

Failure modes. A narrow tour may be mistaken for a comprehensive inventory. Sampling may miss a small hidden stock. Weak chain of custody can make an observed serial untrustworthy. A discrepancy may be a clerical error, not diversion; the response should allow explanation while preserving evidence. The most defensible result is a bounded statement: what was inspected, what was observed, what was not accessible, and how the findings compare with independent records.

An adversarial exercise could place a small number of covered chips in an omitted storage room, swap labels on visible modules, and schedule a legitimate shipment during the visit. Inspectors should identify the hidden stock or explicitly state their inability to rule it out. The exercise should also test whether disputes over customer confidentiality prevent the planned sample from being inspected; formal inspection rights are weak if exceptions consume the key areas.

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